CCSA104 PRACTICAL ASSIGNMENT
MANAGING SECURITY POLICIES

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PART A: PRACTICAL TASKS

Task 1: Explore the Default Policy

Complete observations in SmartConsole:

1. Policy package in use: Standard.

2. Default Cleanup rule, Name: Cleanup Rule.

3. Source: Any.

4. Destination: Any.

5. Services & Applications: Any.

6. Action: Drop.

7. Track: None (Default setting, no logs generated).

8. Install On: Check Point Gateway.

9. Three Global Properties categories:

Firewall

NAT

VPN

(VPN (Virtual Private Network).

10. Policy packages listed under Manage Policies and Layers: Standard

Task 2: Build the Access Control rulebase

Create the objects using the addressing in your lab topology. The following addresses are examples from the assignment; use your actual lab addresses where they differ.

Object name	Type	Example value	Purpose
Internal_LAN	Network	10.1.1.0/24	Staff network
DMZ_Net	Network	172.16.10.0/24	DMZ network
Admin_PC	Host	10.1.1.101	Administrator workstation
Web_Server	Host	172.16.10.20	Public web server
Mail_Server	Host	172.16.10.25	Mail server
External_DNS	Host	192.9.100.10	External DNS server

Configure the Access Control rules in this final order:

No.	Rule name	Source	Destination	Services	Action	Track
1	Admin Access	Admin_PC	Gateway	ssh, https	Accept	Log
2	Stealth Rule	Any	Gateway	Any	Drop	Log
3	Inbound to DMZ	Any	Web_Server, Mail_Server	http, https, smtp	Accept	Log
4	Internal to DMZ	Internal_LAN	DMZ_Net	As required by approved lab scenario	Accept	Log
5	DNS Out	Internal_LAN	External_DNS	dns (UDP and TCP)	Accept	Log
6	Internal Web Out	Internal_LAN	Any	http, https	Accept	Log
7	SMTP Out	Mail_Server	Any	smtp	Accept	Log
8	Cleanup Rule	Any	Any	Any	Drop	Log

Important: Rule 4 is described as being added in Task 5 in the assignment, but its exact service requirements are not specified in the supplied rule table. Confirm the required services with your instructor or Lab 4A guide before configuring it. Do not guess them.

The Cleanup Rule already exists. Do not recreate it. Move it to the bottom if necessary and set Track to Log.

Evidence to insert: full rulebase screenshot showing rule number, name, source, destination, services, action and track.

Task 4: Organise the rulebase with sections

Management and Protection: Rules 1 and 2

DMZ Services: Rules 3 and 4

Outbound Access: Rules 5, 6 and 7

Cleanup: Rule 8

Expand all sections and capture a screenshot. Collapse at least one section and capture a second screenshot. Edit one section title and restore its original wording.

Written response: Rulebase sections are visual organisational headings. They do not change the order in which rules are evaluated, and the section titles themselves are not installed on or enforced by the Security Gateway.

Task 6: Configure Hide NAT

In the Internal_LAN network object, open the NAT page, enable automatic address translation, select Hide, and select the gateway as the hiding address. Publish the changes. Then inspect the NAT rulebase.

Written explanation: This is an automatic NAT rule because it is generated from the NAT settings configured on the Internal_LAN object, rather than being manually entered as a separate rule in the NAT rulebase. Many-to-one translation means that multiple internal devices share one translated public or gateway address when communicating externally. The gateway uses source-port translation (PAT) to distinguish the sessions so that return traffic can be mapped to the correct internal device and connection.

Evidence to insert: Internal_LAN NAT settings screenshot and NAT rulebase screenshot showing the automatically generated rule.

Task 7: Rule order experiment and Installation History

Create TEMP Block Internal with Source Internal_LAN, Destination Any, Services Any, Action Drop and Track Log. First place it above Internal Web Out, publish and install the policy, then repeat test T4. Record the matching rule from the actual log. Move the temporary rule below Internal Web Out, publish and install again, repeat T4, and record the matching rule. Finally delete the temporary rule, publish and install again.

Result with temporary rule above Internal Web Out: TEMP Block Internal (Drop).

Result with temporary rule below Internal Web Out: Internal Web Out (Accept).

Installation History entry containing the temporary rule: Successful policy installation showing TEMP Block Internal.

Three sentence explanation: Security Gateway rules are evaluated from the top of the rulebase downward, and the first matching rule determines the action. A specific rule should therefore be placed before a broader rule when it needs to take precedence. For example, a temporary block rule above Internal Web Out can block traffic that the later web-access rule would otherwise accept.

PART B: THEORY QUESTIONS

Section B1: Multiple Choice

Q1. B, A Security Policy is created in SmartConsole, stored on the Security Management Server and enforced by Security Gateways.

Q2. A, The Positive Control Model (whitelist) allows only explicitly authorised traffic; other traffic is denied.

Q3. C, The Stealth rule should be placed as early as possible, normally immediately after the Management rules, to protect the gateway itself.

Q4. D, Sections are visual divisions for organising the rulebase and do not alter rule enforcement order.

Q5. B, Hide NAT is many-to-one translation: multiple internal addresses share one translated address, with ports used to distinguish connections.

Section B2: Short Answers

Q6. Distinguish between Explicit rules and implied rules. State where implied rules are configured and where they can be placed. What is the Implicit Cleanup Action?

Explicit rules are administrator-created rules entered directly in the Access Control rulebase. Implied rules are system-generated rules controlled through the applicable Global Properties settings in SmartConsole. They are not ordinary manually created rules; their position and behaviour are determined by the relevant implied-rule settings. The Implicit Cleanup Action is the final default action applied to traffic that does not match an applicable rule. Its precise behaviour should be checked in the Global Properties settings of the installed Check Point version.

Q7. Explain why rule order matters in a rulebase. Give one example from your own rulebase.

Rule order matters because the gateway evaluates rules from top to bottom and applies the first matching rule. A broad rule placed too early may match traffic before a more specific rule can be considered. For example, Admin Access should appear before the Stealth Rule so that authorised SSH and HTTPS management traffic from Admin_PC can be accepted before other traffic directed to the gateway is dropped.

Q8. Explain the difference between saving and publishing changes in SmartConsole. What happens to unpublished changes when you install a policy, and why do many organisations publish only during a change window?

Saving preserves work in the administrator's current session or draft, while publishing commits the changes to the Security Management Server so they become part of the management database and available for policy installation. Unpublished changes are not included in an installed policy; only published policy changes can be installed. Organisations often publish during an approved change window to control risk, coordinate testing, minimise disruption and ensure that changes are reviewed and communicated.

Q9. Compare Global Properties with Policy Packages. Give one example of what each controls and say how you open each in SmartConsole.

Global Properties contain system-wide management and security settings that can affect policy behaviour, such as settings for implied rules or other global security options. They are opened from the Applications menu by selecting Global Properties. A Policy Package is a collection of policies and related layers intended for a particular security environment or group of gateways. Policy packages are accessed through Manage policies and layers in SmartConsole. For example, a Global Properties setting may control an implied rule behaviour, while a policy package contains the Access Control rules installed on selected gateways.

Q10. Name the four predefined Security Zones. State two rules about assigning gateway interfaces to zones, and one benefit of using zone based rules.

The four predefined Security Zones are Internal, External, DMZ and VPN. A gateway interface should be assigned to the zone that correctly represents the network connected to that interface. Also, zone assignment should be consistent with the gateway topology and should not create conflicting or ambiguous zone definitions. One benefit of zone-based rules is that they make policies easier to read and manage by allowing administrators to control traffic between network areas according to their security roles.

Section B3: Scenario Questions

Q11. A new administrator inserts a rule with Source Any, Destination Any, Services any and Action Accept above the Stealth rule. Describe two security problems this causes, explain how to correct it, and name the type of rule that is allowed to sit above the Stealth rule and why.

First, the any to Any Accept rule would allow almost all traffic, defeating the Positive Control Model and exposing services that should not be accessible. Second, because it is above the Stealth rule, it could allow unauthorized traffic to reach the gateway itself, weakening protection of the firewall. The correction is to remove the overly permissive rule or replace it with narrowly defined rules specifying approved sources, destinations and services, and place the Stealth rule immediately after the necessary Management rules. Only authorized Management rules should sit above the Stealth rule because administrators need controlled access to manage the gateway.

Q12. After Hide NAT is configured for Internal_LAN, staff report they cannot reach the Internet.

(a) State two benefits of NAT.

1. NAT allows multiple private internal devices to access external networks while sharing a smaller number of public IP addresses. 2. NAT conceals internal IP addressing from external networks, reducing direct exposure of the internal addressing structure.

(b) Describe three checks you would make to find the fault, and explain why each one matters.

Check the Internal_LAN NAT configuration and generated NAT rule. Confirm that Hide NAT is enabled and that the correct gateway address is selected; an incorrect translation setting can prevent return traffic from reaching internal hosts.

Check the gateway's routing, default route and external interface connectivity. The gateway must have a valid path to the Internet for translated traffic to leave the network and for replies to return.

Check the Access Control policy and traffic logs for the affected client. Confirm that the required outbound services are accepted and inspect logs for drops or NAT-related errors; a policy denial or other rule/configuration issue may be responsible.